

ON THE CONGRUENCES $2^n \equiv -1$ AND $2^n \equiv -3 \pmod{n+1}$

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ABSTRACT. We study which integers m satisfy $m \mid 2^{m-1} + 3$, equivalently which n satisfy $n+1 \mid 2^n + 3$ (OEIS A245728). Only two are known— $61 \cdot 228806497$ and $67 \cdot 44210291368986343$, each a product of two primes—and none lies below 10^{15} . Our central result ties the question to the prime factorisation of m : a squarefree $m = p_1 \cdots p_\omega$ is a solution if and only if, for every i , -3 is a power of 2 modulo p_i and $m \equiv k_i + 1 \pmod{d_i}$, where $d_i = \text{ord}_{p_i}(2)$ and $2^{k_i} \equiv -3 \pmod{p_i}$. Equivalently, the largest prime factor of m must divide $2^{P-1} + 3$ with P the product of the remaining primes, so locating solutions reduces to factoring a single, rapidly growing integer. These local conditions exclude a positive proportion of primes as factors and cannot be sharpened by prime-power moduli; from them we show that every solution is squarefree at each prime below 1093, and that the only two-prime solutions with smaller prime at most 499 are the two above. Heuristics suggest that this set is infinite (§12)—though extraordinarily sparse, dominated by many-prime m beyond any feasible factorisation, which is precisely why only two-prime examples have been found. By contrast, $m \mid 2^{m-1} + 1$ has no solutions, by a short 2-adic argument.

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1. INTRODUCTION

For an integer a , write $S_a = \{n \geq 1 : (n+1) \mid 2^n - a\}$. Setting $m = n+1$, membership $n \in S_a$ is equivalent to

$$m \mid 2^{m-1} - a, \quad m \geq 2. \quad (1)$$

This note concerns $a = -1$ and $a = -3$. We prove $S_{-1} = \emptyset$ and assemble, both structurally and computationally, what is presently known about S_{-3} .

Notation. For an odd prime p with $p \nmid 2$, write $d_p = \text{ord}_p(2)$, the multiplicative order of 2 in $\mathbb{F}_p^\times$. For $a \in \mathbb{Z}$ with $\gcd(a, p) = 1$ and $a \in \langle 2 \rangle \subseteq \mathbb{F}_p^\times$, write $\log_2 a$ for the smallest non-negative integer k with $2^k \equiv a \pmod{p}$. We use $\widetilde{\log}_2 a$ analogously for the discrete logarithm in $(\mathbb{Z}/p^2)^\times$. The 2-adic valuation of an integer N is $v_2(N)$.

2. THE CASE $a = -1$

Theorem 1. *There is no integer $m > 1$ with $m \mid 2^{m-1} + 1$. Equivalently, $S_{-1} = \emptyset$.*

Proof. Suppose such an m exists. The integer $2^{m-1} + 1$ is odd, so m is odd. Fix any prime $p \mid m$ and set $d = d_p$. From $p \mid 2^{m-1} + 1$,

$$2^{m-1} \equiv -1 \pmod{p}, \quad 2^{2(m-1)} \equiv 1 \pmod{p}.$$

Hence $d \mid 2(m-1)$ but $d \nmid m-1$ (since p is odd, $-1 \not\equiv 1 \pmod{p}$). It follows that $d/\gcd(d, m-1) = 2$, so $\gcd(d, m-1) = d/2$. Comparing 2-adic valuations, $v_2(d) - 1 = v_2(\gcd(d, m-1)) \leq v_2(m-1)$ on the one hand, while $d \nmid m-1$ together with $d \mid 2(m-1)$ forces $v_2(d) > v_2(m-1)$ on the other. Therefore

$$v_2(d) = v_2(m-1) + 1. \quad (2)$$

Now $d \mid p-1$, so $v_2(p-1) \geq v_2(d) = v_2(m-1) + 1$. Setting $\nu = v_2(m-1) + 1$, we have shown $p \equiv 1 \pmod{2^\nu}$ for every prime $p \mid m$. Since m is a product of primes in $1 + 2^\nu \mathbb{Z}$, we get $m \equiv 1 \pmod{2^\nu}$, i.e. $2^\nu \mid m-1$. But $\nu = v_2(m-1) + 1$, contradicting the definition of $v_2(m-1)$. $\square$

Remark 1. The proof gives slightly more: for any $m \geq 2$ and any odd a with $\gcd(a, m) = 1$ and $a \equiv -1 \pmod{p}$ for every $p \mid m$, the same 2-adic argument applies. The obstruction is genuinely about square roots of unity, and this is exactly why it does not transfer to $a = -3$: -3 is not a root of unity modulo a typical prime, so no global multiplicative collapse occurs.

3. THE CASE $a = -3$: ELEMENTARY RESTRICTIONS

We turn to (1) with $a = -3$, that is,

$$m \mid 2^{m-1} + 3. \quad (\star)$$

Proposition 1. *Let $m \geq 2$ satisfy $(\star)$. Then:*

- (i) m is odd, equivalently $n = m-1$ is even;
- (ii) $3 \nmid m$;
- (iii) m is composite.

Proof. (i) $2^{m-1} + 3$ is odd, so m is odd. (ii) If $3 \mid m$ then $3 \mid 2^{m-1} + 3$, hence $3 \mid 2^{m-1}$, impossible. (iii) If m is prime, Fermat's little theorem gives $2^{m-1} \equiv 1 \pmod{m}$, so $m \mid 4$, forcing $m \in \{2\}$, contradicting (i). $\square$

4. THE KNOWN SOLUTIONS

The integers n satisfying $(n+1) \mid 2^n + 3$ form OEIS sequence A245728 [1]. Only two terms are known:

$$m_1 = n_1 + 1 = 13957196317 = 61 \cdot 228806497,$$

$$m_2 = n_2 + 1 = 2962089521722084981 = 67 \cdot 44210291368986343,$$

and no further n exists with $n+1 < 10^{15}$. Both factorisations are products of two distinct primes; in each case the smaller prime is small (61 or 67) and the cofactor is large. Both were verified directly to satisfy $(\star)$, and all four factors are prime. The second solution exceeds 10^{18} and was found structurally rather than by enumeration; see Section 11.

5. ADMISSIBILITY: A SIEVE ON PRIME FACTORS

The next lemma gives a strong necessary condition that every prime divisor of m must satisfy; it underlies all subsequent analysis.

Lemma 1 (Local condition at a prime divisor). *Let $m \geq 5$ satisfy $(\star)$, let p be a prime dividing m , and write $m = p^a t$ with $\gcd(p, t) = 1$. Then $p \notin \{2, 3\}$, the element -3 lies in $\langle 2 \rangle \subseteq \mathbb{F}_p^\times$, and, writing $d = d_p$ and $k = \log_2(-3) \pmod{p}$,*

$$t \equiv k + 1 \pmod{d}. \quad (3)$$

Proof. By Proposition 1, p is odd and $p \neq 3$, so $\gcd(p, 6) = 1$. Since $d \mid p-1$, $p \equiv 1 \pmod{d}$, hence $p^a \equiv 1 \pmod{d}$ and $m \equiv t \pmod{d}$. Therefore $m-1 \equiv t-1 \pmod{d}$ and $2^{m-1} \equiv 2^{t-1} \pmod{p}$. Since $p \mid 2^{m-1} + 3$, this gives $2^{t-1} \equiv -3 \pmod{p}$, which both shows $-3 \in \langle 2 \rangle$ and yields (3). $\square$

Definition 1. A prime $p \notin \{2, 3\}$ is *admissible* if $-3 \in \langle 2 \rangle \subseteq \mathbb{F}_p^\times$ and, writing $d = d_p$ and $k = \log_2(-3)$:

- (A) if d is even, then k is even;
- (B) if $3 \mid d$, then $k \not\equiv 2 \pmod{3}$.

Theorem 2. *Every prime divisor of any m satisfying $(\star)$ is admissible.*

Proof. Fix $p \mid m$ and use the notation of Lemma 1, with $t = m/p^{v_p(m)}$. Since m is odd, t is odd; since $3 \nmid m$, $3 \nmid t$. Reduce (3) modulo 2 and modulo 3. If d is even, then $t \pmod{d}$ has the same parity as t , namely odd, so $k+1$ is odd, i.e. k is even. If $3 \mid d$, then $t \pmod{d}$ has the same residue modulo 3 as t , which is 1 or 2 but not 0, so $k+1 \not\equiv 0 \pmod{3}$, i.e. $k \not\equiv 2 \pmod{3}$. $\square$

Numerical effect of the sieve. Among the primes $p < 200$ for which $-3 \in \langle 2 \rangle$ in $\mathbb{F}_p^\times$ (a necessary condition without which p cannot divide any m), Theorem 2 eliminates substantially more than half. Table 1 records the data; admissible primes are marked $\checkmark$. The admissible primes below 200 are

$$13, 19, 61, 67, 79, 103, 193, 199.$$

The two known solutions use the admissible primes 61 and 67 together with admissible large cofactors (one verifies directly that 228806497 and 44210291368986343 are admissible). Condition (A) eliminates roughly half of the eligible primes (those with d even and k odd, which by equidistribution is half), and condition (B) eliminates a further third of those with $3 \mid d$. Empirically, over primes below 2000, about 56% are -3 -eligible and the sieve removes about 61.5% of those, leaving roughly 21.6% of all primes admissible. Thus admissible primes form a sparse but *positive-proportion* subsequence (conjecturally of positive density, by an Artin-type heuristic); in particular the supply of candidate small factors is infinite, which already signals that no finite search can be conclusive.

p	d_p	k	(A) k even if $2 \mid d$	(B) $k \not\equiv 2 \pmod{3}$ if $3 \mid d$	adm.
5	4	1	fail	—	×
7	3	2	—	fail	×
11	10	3	fail	—	×
13	12	10	ok	ok	✓
19	18	4	ok	ok	✓
29	28	19	fail	—	×
37	36	8	ok	fail	×
53	52	43	fail	—	×
59	58	21	fail	—	×
61	60	36	ok	ok	✓
67	66	6	ok	ok	✓
79	39	10	—	ok	✓
83	82	31	fail	—	×
97	48	43	fail	ok	×
101	100	19	fail	—	×
103	51	9	—	ok	✓
107	106	17	fail	—	×
131	130	7	fail	—	×
139	138	110	ok	fail	×
149	148	13	fail	—	×
163	162	20	ok	fail	×
173	172	113	fail	—	×
179	178	19	fail	—	×
181	180	146	ok	fail	×
193	96	90	ok	ok	✓
197	196	83	fail	—	×
199	99	85	—	ok	✓

TABLE 1. Primes $p < 200$ with $-3 \in \langle 2 \rangle \pmod{p}$. Column (A) is vacuous when d_p is odd; column (B) is vacuous when $3 \nmid d_p$.

6. INTERLOCKING CONGRUENCES

The local condition (3) globalises by the Chinese Remainder Theorem.

Proposition 2. *Let m satisfy $(\star)$ with prime factorisation $m = \prod_{i=1}^r p_i^{a_i}$. For each i , set $d_i = d_{p_i}$ and $k_i = \log_2(-3) \pmod{p_i}$, and put $t_i = m/p_i^{a_i}$. Then $t_i \equiv k_i + 1 \pmod{d_i}$ for every i , and consequently*

$$m \equiv p_i^{a_i}(k_i + 1) \pmod{p_i^{a_i}d_i}$$

for every i . By CRT, m lies in a single residue class modulo $\text{lcm}_i(p_i^{a_i}d_i)$.

Proof. The first statement is Lemma 1. For the second, $m = p_i^{a_i}t_i$ and $t_i \equiv k_i + 1 \pmod{d_i}$, so $m \equiv p_i^{a_i}(k_i + 1) \pmod{p_i^{a_i}d_i}$. $\square$

For both known solutions one verifies Proposition 2 explicitly:

$$m_1 = 61 \cdot 228806497 : \quad 228806497 \pmod{60} = 37 = k_{61} + 1,$$

$$m_2 = 67 \cdot 44210291368986343 : \quad 44210291368986343 \pmod{66} = 7 = k_{67} + 1.$$

7. TWO-PRIME STRUCTURE

Both known m are products of two distinct primes. Specialising Proposition 2 gives a tight description of all such solutions, and the next remark shows that one half of the description is automatic.

Corollary 1. *Suppose $m = pq$ with primes $p < q$, $p, q \neq 2, 3$. Then m satisfies $(\star)$ if and only if both*

$$q \equiv k_p + 1 \pmod{d_p}, \quad p \equiv k_q + 1 \pmod{d_q} \quad (4)$$

hold, where d_p, d_q are the orders of 2 and k_p, k_q the discrete logarithms of -3 to base 2 in $\mathbb{F}_p^\times$ and $\mathbb{F}_q^\times$ respectively. In particular q is a prime divisor of $2^{p-1} + 3$ and p is a prime divisor of $2^{q-1} + 3$.

Remark 2 (One-sidedness). The two conditions in (4) are not independent once q is drawn from the prime factors of $N_p := 2^{p-1} + 3$. Indeed, if $q \mid N_p$ then $2^{p-1} \equiv -3 \pmod{q}$, and a short computation with Fermat's little theorem shows

$$2^{pq-1} \equiv (2^{p-1})^q \cdot 2^{q-1} \equiv (-3)^q \cdot 1 \equiv -3 \pmod{q},$$

so the congruence modulo q holds automatically. Equivalently, the second condition $p \equiv k_q + 1 \pmod{d_q}$ holds for *every* prime factor q of N_p . Hence

$$m = pq \ (p < q) \text{ solves } (\star) \iff q \mid 2^{p-1} + 3 \text{ and } q \equiv k_p + 1 \pmod{d_p}.$$

This makes the two-prime problem for a fixed smaller prime p entirely a question about the prime factorisation of the single integer N_p , together with one residue test modulo d_p ($< p$).

This is the basis of an efficient algorithm: for each admissible small prime p , factor $N_p = 2^{p-1} + 3$, and for each prime factor $q > p$ test whether $q \equiv k_p + 1 \pmod{d_p}$.

8. SOLUTIONS WITH MORE PRIME FACTORS

Corollary 1 is the case $\omega(m) = 2$. The same reduction works for any number of prime factors; it both clarifies the structure of hypothetical longer solutions and explains why only two-prime ones are known. Throughout, $m = p_1 p_2 \cdots p_\omega$ is squarefree with $p_1 < p_2 < \cdots < p_\omega$ distinct primes, $d_i = d_{p_i}$ and $k_i = \log_2(-3) \pmod{p_i}$. (By Section 9 every solution is squarefree at all primes below 1093, so a repeated prime factor, if one ever occurs, must be ≥ 1093 ; we return to this at the end.)

Proposition 3 (Squarefree solutions of any length). *Let $m = p_1 \cdots p_\omega$ be squarefree with each $p_i \notin \{2, 3\}$. Then $m \mid 2^{m-1} + 3$ if and only if, for every i ,*

$$-3 \in \langle 2 \rangle \subseteq \mathbb{F}_{p_i}^\times \quad \text{and} \quad m \equiv k_i + 1 \pmod{d_i},$$

the latter being equivalent to $\prod_{j \neq i} p_j \equiv k_i + 1 \pmod{d_i}$. In particular every p_i is admissible.

Proof. Since m is squarefree, the CRT gives $m \mid 2^{m-1} + 3$ iff $p_i \mid 2^{m-1} + 3$ for every i . For fixed i , $2^{m-1} \equiv -3 \pmod{p_i}$ holds iff $-3 \in \langle 2 \rangle$ in $\mathbb{F}_{p_i}^\times$ and $m - 1 \equiv k_i \pmod{d_i}$. As $d_i \mid p_i - 1$ we have $p_i \equiv 1 \pmod{d_i}$, hence $m \equiv \prod_{j \neq i} p_j \pmod{d_i}$, giving the reformulation. Conditions (A),(B) of Definition 1 follow as in Theorem 2. $\square$

Thus a squarefree solution is exactly a set of admissible primes whose product lands simultaneously in the prescribed class $k_i + 1$ modulo each d_i . By CRT this confines m to one residue class modulo $\text{lcm}_i d_i$ (Proposition 2); the difficulty is that the unknown m is forced to be the *product* of the very primes defining the moduli. The two-prime reduction of Remark 2 generalises verbatim: the condition at the largest prime is a divisibility, and the remaining ones are residue tests.

Corollary 2 (Recursive criterion). *Let $p_1 < \dots < p_\omega$ be admissible primes, put $P = p_1 \cdots p_{\omega-1} = m/p_\omega$, and write $N_P = 2^{P-1} + 3$. Then $m = Pp_\omega$ solves $(\star)$ if and only if*

$$p_\omega \mid N_P \quad \text{and} \quad m \equiv k_i + 1 \pmod{d_i} \quad (1 \leq i \leq \omega - 1).$$

The congruence $m \equiv k_\omega + 1 \pmod{d_\omega}$ is automatic once $p_\omega \mid N_P$.

Proof. The congruence at p_ω reads $P \equiv k_\omega + 1 \pmod{d_\omega}$ (since $m \equiv P \pmod{d_\omega}$), i.e. $2^{P-1} \equiv -3 \pmod{p_\omega}$, i.e. $p_\omega \mid N_P$; this holds for every prime factor of N_P . The remaining $\omega - 1$ congruences are Proposition 3. $\square$

This is an algorithm: to find every ω -prime solution extending a fixed tuple $(p_1, \dots, p_{\omega-1})$, factor the single integer $N_P = 2^{P-1} + 3$ and test each prime factor $p_\omega > p_{\omega-1}$ against the $\omega - 1$ residue conditions modulo $d_1, \dots, d_{\omega-1}$. Two points deserve emphasis. First, the smaller primes need *not* themselves form a solution: solutions do not nest, so one cannot grow a long solution by extending a short one. Second, everything reduces to factoring N_P , and that is precisely where the method fails.

8.1. The computational barrier. The integer $N_P = 2^{P-1} + 3$ has about $0.301 P$ decimal digits, and $P = p_1 \cdots p_{\omega-1}$ is a product of admissible primes, hence grows super-exponentially with ω . Even the smallest admissible primes 13, 19, 61, 67, $\dots$, which minimise P , give the sizes of Table 2.

ω	minimal $P = p_1 \cdots p_{\omega-1}$	digits of N_P	feasibility
2	13	4	routine (broad search)
3	$13 \cdot 19 = 247$	75	at the edge (one tuple)
4	$13 \cdot 19 \cdot 61 = 15067$	$\approx 4,500$	infeasible
5	$13 \cdot 19 \cdot 61 \cdot 67 = 1,009,489$	$\approx 3 \times 10^5$	infeasible

TABLE 2. Smallest integer $N_P = 2^{P-1} + 3$ that must be factored to search for an ω -prime solution.

So two prime factors is the only regime open to a broad search, three is reachable for a single tuple, and four or more is permanently beyond factorisation. Concretely, for three primes only the smallest admissible pair $(p_1, p_2) = (13, 19)$ keeps N_P tractable:

$$N_{247} = 2^{246} + 3 = 307 \cdot 41047 \cdot r, \quad r \text{ a 67-digit prime,}$$

and none of $13 \cdot 19 \cdot p_3$ with $p_3 \in \{307, 41047, r\}$ satisfies the two residue conditions, so there is no three-prime solution with smaller pair (13, 19). Every other pair already pushes N_P past 200 digits, so a complete three-prime search—let alone $\omega \geq 4$ —is not currently possible.

8.2. How many solutions, and how large? Because the multi-prime regime is computationally invisible, the only guide is heuristic. Modelling $2^{m-1} + 3 \pmod{m}$ as equidistributed makes a squarefree m with all prime factors admissible a solution with “probability” about $1/m$; when $d_i \approx p_i$ this agrees with the structural estimate $\prod_i 1/d_i$. Summing over candidates with exactly ω prime factors gives, for the number of ω -prime solutions up to X ,

$$E_\omega(X) \approx \frac{1}{\omega!} \left(\sum_{\substack{p \leq X \\ \text{admissible}}} \frac{1}{p} \right)^\omega \approx \frac{(\delta \log \log X)^\omega}{\omega!},$$

where δ is the (logarithmic) density of admissible primes, empirically $\delta \approx 0.15$. Hence

$$\sum_{\omega \geq 2} E_\omega(X) \approx e^{\delta \log \log X} - 1 - \delta \log \log X = (\log X)^\delta - 1 - \delta \log \log X \xrightarrow{X \rightarrow \infty} \infty.$$

Three conclusions follow, of which the third is the point. (i) The total diverges, so the heuristic predicts *infinitely many* solutions—but only as a minuscule power $(\log X)^\delta$ of $\log X$, so they are

astronomically sparse. (ii) The estimate is of the right order: at the scale of the known terms ($X \approx m_2$) it gives $\sum_{\omega \geq 2} E_\omega$ of order 10^{-1} , so finding two solutions is consistent with the model to within its (considerable) crudeness; the structural refinement $\prod_i 1/d_i$, which exceeds $1/m$ when some order d_i is small, raises the estimate toward the observed count. (iii) At every reachable scale $\delta \log \log X < 1$, so E_ω decreases with ω : two-prime solutions are heuristically the most common, and a solution with three or more prime factors is not expected until X is of the order $\exp(\exp(c/\delta))$ —a number with thousands of digits or more. The two known solutions are two-prime both because two-prime solutions genuinely dominate at small heights and because, even were a longer solution to lie in range, exhibiting it would require factoring an N_P of thousands of digits. Longer solutions are thus doubly hidden: too rare to meet in any searchable range, and walled off by factorisation when they do occur.

8.3. Non-squarefree m . If $p^a \parallel m$ with $a \geq 2$, Proposition 3 must be read modulo p^a : the condition at p becomes $2^{m-1} \equiv -3 \pmod{p^a}$, governed by $\text{ord}_{p^a}(2)$ and the discrete logarithm of -3 there, in the manner of Lemma 2. Corollary 3 forbids $a \geq 2$ for every prime $p < 1093$, so a non-squarefree solution would have to involve a repeated prime ≥ 1093 ; none is known, and the heuristic above, dominated by squarefree m , is unaffected.

9. SQUAREFREENESS AT SMALL PRIMES

Both known solutions are squarefree. We now prove that this is forced at every non-Wieferich prime, modulo a single discrete-logarithm coincidence that we then verify never occurs below 1093. Recall that p is a *Wieferich prime* (base 2) if $2^{p-1} \equiv 1 \pmod{p^2}$; the only such primes below 4×10^3 are 1093 and 3511. For p non-Wieferich one has $\text{ord}_{p^2}(2) = p d_p$ (if $2^{d_p} \equiv 1 \pmod{p^2}$ then $2^{p-1} \equiv 1 \pmod{p^2}$, since $d_p \mid p-1$; the contrapositive gives the order $p d_p$).

Lemma 2 (Squarefreeness at a non-Wieferich prime). *Let m satisfy $(\star)$ and let p be a prime with $p^2 \mid m$. Suppose p is not a Wieferich prime base 2, so $D := \text{ord}_{p^2}(2) = p d_p$. Then $-3 \in \langle 2 \rangle \subseteq (\mathbb{Z}/p^2)^\times$ and, writing $\tilde{k}_p := \widetilde{\log_2(-3)}$ for its discrete logarithm modulo p^2 ,*

$$\tilde{k}_p \equiv -1 \pmod{p}.$$

Proof. From $p^2 \mid m \mid 2^{m-1} + 3$ we get $2^{m-1} \equiv -3 \pmod{p^2}$, so $-3 \in \langle 2 \rangle$ modulo p^2 and

$$m-1 \equiv \tilde{k}_p \pmod{D}.$$

Since p is non-Wieferich, $D = p d_p$, so $p \mid D$ and we may reduce the previous congruence modulo p :

$$m-1 \equiv \tilde{k}_p \pmod{p}.$$

But $p^2 \mid m$ gives $m \equiv 0 \pmod{p}$, hence $m-1 \equiv -1 \pmod{p}$. Therefore $\tilde{k}_p \equiv -1 \pmod{p}$. $\square$

Remark 3. The mechanism here differs from that of the sieve. Conditions (A),(B) of Definition 1 arise from the cofactor $t = m/p^a$ being odd and coprime to 3, which constrains t only modulo 2 and 3; the classes modulo 4, 9, ... remain free (Section 10). In Lemma 2 the modulus is the prime p itself, and the order $D = p d_p$ ties the residue $m \bmod p$, which is 0 since $p^2 \mid m$, to $\tilde{k}_p \bmod p$. This is a genuine constraint rather than a free residue class.

Corollary 3. *For every admissible prime $p < 1093$ one has $\tilde{k}_p \not\equiv -1 \pmod{p}$. Consequently no solution m of $(\star)$ is divisible by p^2 for any prime $p < 1093$; that is, every $m \in S_{-3}$ is squarefree at all primes below 1093.*

Proof. If $p^2 \mid m$ then $p \mid m$, so p is admissible by Theorem 2, and every admissible $p < 1093$ is non-Wieferich. By Lemma 2 (and noting that if $-3 \notin \langle 2 \rangle$ modulo p^2 then $p^2 \nmid m$ outright), divisibility by p^2 would force $\tilde{k}_p \equiv -1 \pmod{p}$. A direct computation of the discrete logarithm $\tilde{k}_p$ for each of the 39 admissible primes $p < 1093$ shows $\tilde{k}_p \not\equiv -1 \pmod{p}$ in every case, a contradiction. $\square$

p	d_p	$\tilde{k}_p \bmod p$	$-1 \bmod p$
13	12	7	12
19	18	6	18
61	60	19	60
67	66	35	66
79	39	7	78
103	51	66	102
193	96	7	192
199	99	35	198
211	210	186	210

TABLE 3. For small admissible primes, the reduced discrete logarithm $\tilde{k}_p = \widetilde{\log_2(-3) \bmod p^2}$ taken modulo p , against $-1 \bmod p$. Equality (which would permit $p^2 \mid m$) never occurs; verified for all admissible $p < 1093$.

Table 3 records $\tilde{k}_p \bmod p$ for the small admissible primes; the value $-1 \bmod p = p - 1$ never appears.

A full proof of squarefreeness would also require a uniform treatment of large primes and a separate analysis of the Wieferich primes 1093 and 3511; the present argument settles only the primes below 1093.

10. NO PRIME-POWER SHARPENING OF THE SIEVE

It is natural to ask whether replacing the moduli 2, 3 in conditions (A),(B) by higher prime powers 4, 8, ... or 9, 27, ... yields additional exclusions. It does not.

Proposition 4. *The only constraints on $t = m/p^a$ available in isolation are t odd and $3 \nmid t$. Reducing the local congruence (3) modulo any power 2^e or 3^f produces no exclusion beyond conditions (A) and (B).*

Proof. Suppose $2^e \mid d$. Then (3) forces $t \equiv k + 1 \pmod{2^e}$. The constraint t odd permits exactly the odd residues modulo 2^e , and the forced value $k + 1$ fails to be odd precisely when $k + 1$ is even, i.e. when k is odd. This is condition (A) (which already applies as soon as $2 \mid d$). Higher e add nothing, since oddness is a condition modulo 2 only. Similarly, if $3^f \mid d$ then $t \equiv k + 1 \pmod{3^f}$, and $3 \nmid t$ permits exactly the residues coprime to 3; the forced value $k + 1$ fails to be coprime to 3 precisely when $3 \mid k + 1$, i.e. $k \equiv 2 \pmod{3}$, which is condition (B). Higher f add nothing. $\square$

We verified computationally that no admissible prime below 3000 is removed by any mod-4, mod-8, mod-9, or mod-27 refinement, consistent with Proposition 4. Any genuine sharpening of the sieve must therefore come from *interactions between distinct prime factors*, i.e. from the interlocking congruences of Proposition 2, rather than from a single local modulus.

11. COMPUTATIONAL EVIDENCE

11.1. Direct search. The bound $m \leq 10^{15}$ is the range exhaustively verified in OEIS A245728 [1]; the only solution there is m_1 . An independent brute-force test over all odd $m \leq 10^7$ with $3 \nmid m$ reproduces no solution in that range, agreeing with the database.

11.2. Two-prime search via factorisation of $2^{p-1} + 3$. By Remark 2, for each admissible smaller prime p it suffices to factor $N_p = 2^{p-1} + 3$ and test the prime factors $q > p$ against $q \equiv k_p + 1 \pmod{d_p}$. We carried this out for every admissible $p \leq 499$:

smaller prime p	$N_p = 2^{p-1} + 3$ status	solution
13, 19	full	none
61	full	$m_1 = 61 \cdot 228806497$
67	full	$m_2 = 67 \cdot 44210291368986343$
79, 103, 193, 199, 211, 271, 307, 373, 463, 499	full	none
367, 439, 487	partial (109–147-digit cofactor)	open

TABLE 4. Two-prime search over admissible smaller primes $p \leq 499$. By Theorem 2 only admissible primes can occur as a smaller factor, so non-admissible primes require no factorisation.

- For $p \in \{13, 19, 61, 67, 79, 103, 193, 199, 211, 271, 307, 373, 463, 499\}$ the number N_p is fully factored (by trial division, Pollard ρ , $p-1$, and ECM). The complete list of two-prime solutions with smaller prime in this set is exactly $\{m_1, m_2\}$, arising at $p = 61$ and $p = 67$.
- For $p \in \{367, 439, 487\}$ a composite cofactor of 109, 132, and 147 decimal digits respectively remains unfactored; in each case all prime factors found so far yield no solution, and (each residual cofactor having been checked to be composite, not a single large prime) the possibility of a further two-prime m involving these p is left open.

A search for solutions with three or more prime factors is governed by the recursive criterion of Section 8; only the single tuple (13, 19) is within reach, and it yields no three-prime solution.

12. IS S_{-3} FINITE?

This is the heart of “fully solving” the problem, and it is open. The available heuristics—a structural estimate for two-prime solutions and a global density argument—point the same way, toward infinitude; we give both and then say what a proof would require.

The two-prime estimate. For admissible p , the number $N_p = 2^{p-1} + 3$ has about $\log \log N_p \asymp \log p$ distinct prime factors, and a given one lies in the class required modulo d_p with “probability” about $1/d_p$. The expected number of two-prime solutions with smaller prime below X is thus of order $\sum_{p < X, \text{ adm}} (\log p)/d_p$, and this *diverges*. Since $d_p \mid p-1$ we have $d_p < p$, hence $(\log p)/d_p > (\log p)/p$; assuming, as the data strongly support, that the admissible primes have positive density δ , Mertens’ first theorem gives

$$\sum_{\substack{p < X \\ \text{admissible}}} \frac{\log p}{d_p} \geq \sum_{\substack{p < X \\ \text{admissible}}} \frac{\log p}{p} \sim \delta \log X \xrightarrow{X \rightarrow \infty} \infty.$$

The factor $\log p$ in the numerator is what fixes the rate at $\log X$: the bare sum $\sum 1/d_p$, lower-bounded by $\sum 1/p$, diverges only at the slower rate $\log \log X$ (numerically $\sum_{p < X, \text{ adm}} 1/d_p$ creeps from about 0.24 at $X = 200$ to 0.53 at $X = 5 \cdot 10^4$). So the two-prime heuristic predicts *infinitely many* two-prime solutions, not finiteness. It is moreover well calibrated: $\sum_{p < X, \text{ adm}} (\log p)/d_p$ is about 1.7 at $X = 2000$ and 2.4 at $X = 10^4$, consistent with the two known solutions, whose smaller primes are 61 and 67.

A global density heuristic. Model $2^{m-1} + 3 \pmod m$ as roughly equidistributed; then m is a solution with “probability” about $1/m$, and $\sum_{m \leq X} 1/m \sim \log X \rightarrow \infty$. A divergent expectation predicts *infinitely many* solutions. This is precisely the (correct) heuristic for the closely analogous Fermat-pseudoprime problem $m \mid 2^{m-1} - 2$, for which infinitely many solutions are known (Erdős). The inhomogeneous shift (-3 in place of -1) changes the combinatorics of how prime factors glue together but not the divergence of the expectation.

Both estimates thus point the same way. The two-prime sum already diverges, and the global heuristic is dominated by solutions with *many* prime factors, whose expected distribution (Section 8) also diverges; together they favour an infinite S_{-3} whose elements beyond m_2 are many-prime

integers far beyond any search. On this view the two known terms being two-prime reflects the limits of factorisation, not the true distribution of solutions.

A proof of *finiteness* would need an upper bound, uniform in p , on the number of prime factors of $2^{p-1} + 3$ lying in prescribed arithmetic progressions, which is well beyond current analytic number theory. A proof of *infinitude* would need a Carmichael-style construction producing solutions to order; the homogeneous case $a = -1$ admits the relevant root-of-unity structure (and is settled in the opposite direction, $S_{-1} = \emptyset$), but for $a = -3$ the varying targets k_q obstruct the standard construction and none is known. Both directions are presently out of reach.

13. OPEN QUESTIONS

1. *Is S_{-3} finite?* This remains open; the heuristics of Section 12 point toward an infinite, though computationally unreachable, solution set.
2. *Can the admissibility sieve be sharpened locally?* No: Proposition 4 shows that prime-power moduli yield nothing further, so any improvement must be global.
3. *Is every $m \in S_{-3}$ squarefree?* This is open in general; Corollary 3 settles it at every prime below 1093, and a uniform treatment of large primes, together with a separate analysis of the Wieferich primes 1093, 3511, would be needed for the full statement.
4. *The same questions for $a \in \{-5, -7, \dots\}$.* The proof of Theorem 1 uses critically that -1 is a root of unity; for general a the 2-adic argument fails and the qualitative behaviour of S_a is unclear.

APPENDIX A. COMPUTATIONAL PROGRAMS

The programs used for the computations reported here—the direct search, the two-prime factoring search via $N_p = 2^{p-1} + 3$, and the supporting order and discrete-logarithm routines—are available at <https://github.com/RobinCodes/Projects/tree/main/Divisibility%20problem>.

The factorisations of Table 4, the discrete-logarithm verification underlying Corollary 3, and the sieve data of Table 1 were carried out with `sympy`.

APPENDIX B. NOTE ON THE USE OF ARTIFICIAL INTELLIGENCE

This work was prepared with substantial assistance from an artificial-intelligence system, and most of the results presented here—the admissibility sieve, the interlocking-congruence description, the two-prime reduction, the squarefreeness analysis, and the computations of Section 11—were created with such assistance. We do not specify the particular system used; readers seeking further detail about the methods are invited to contact the authors: Robin Rovenszky (`robincodes` on Discord) and notxxdog (`.notxxdog` on Discord).

No result here is accepted on the basis of machine assertion. Every definition, lemma, and proof is stated so as to be checkable by hand, and every computational claim is reproducible by re-running the programs of Appendix A.

REFERENCES

- [1] OEIS Foundation Inc. (2024), Entry A245728 in *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A245728>.